1 Proof Template: Showing that a real number is the supremum of a set

Proposition 1.1. Define a subset $S \subset \mathbb{R}$ by blah, blah, blah, and let u^* be the real number blah. Then $u^* = \sup S$.

Proof.

(i) We first show that u^* is an upper bound for S. Let $s \in S$. Then blah, blah, blah. It follows that $s \leq u^*$.

(ii) Next we show that u^* is the least upper bound for S. Suppose that u is an upper bound for S. Then blah, blah, blah. It follows that $u^* \leq u$.

Alternatively, we show that u^* is the least upper bound for S as follows: suppose $u < u^*$. Then u is not an upper bound for S since blah, blah, blah.

□

Example 1.1. Let $S = (-\infty, 1)$. Then $\sup(S) = 1$.

Proof. We will use the above template to prove that $\sup(S) = 1$.

(i) We first show that 1 is an upper bound for S. This follows immediately from the definition

$$S = (-\infty, 1) := \{x \in \mathbb{R} : x < 1\}.$$

So if $s \in S$, then $s < 1$.

(ii) Next, we show that 1 is the least upper bound for S. Suppose $x < 1$. Then the midpoint between x and 1 is given by $(x+1)/2$. This midpoint satisfies $x < \frac{(x+1)}{2} < 1$, so it is greater than x and less than 1. Since it is less than 1, we have $\frac{x+1}{2} \in S$, by definition of the set S. It follows that x is not an upper bound for S, since there exists an element $\frac{(x+1)}{2} \in S$ with $\frac{(x+1)}{2} > x$.

We have shown that 1 is an upper bound for S and anything smaller than 1 is not an upper bound. So $1 = \sup(S)$. □

2 Characterization of the Supremum

The following proposition gives a very useful ε-characterization of the supremum of a nonempty upper bounded subset of $\mathbb{R}$.

Proposition 2.1. Let S be a nonempty upper bounded subset of $\mathbb{R}$ and let $u^* \in \mathbb{R}$. Then $u^* = \sup S$ if and only if:

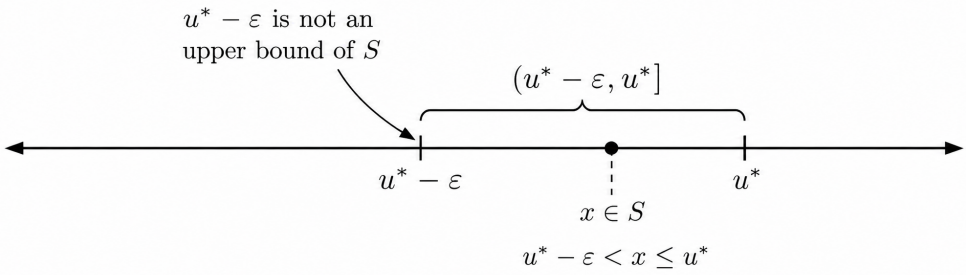
(a) u^* is an upper bound of S;

(b) $\forall \varepsilon > 0$ (for every $\varepsilon > 0$) $\exists x \in S$ (there exists $x \in S$) such that

$$x > u^* - \varepsilon.$$

Geometric meaning. Since u^* is an upper bound, every point of S lies at or to the left of u^*. Condition (b) says that no matter how small $\varepsilon > 0$ is, the interval immediately to the left of u^* contains a point of S; equivalently,

$$S \cap (u^* - \varepsilon, u^*] \neq \varnothing.$$



Proof. Suppose first that $u^* = \sup S$. By the definition of supremum, (a) holds.
For (b), let $\varepsilon > 0$. If for every $x \in S$ we had $x \leq u^* - \varepsilon$, then $u^* - \varepsilon$ would be an upper bound of S. By the definition of supremum we would then have

$$u^* \leq u^* - \varepsilon,$$

that is, $\varepsilon \leq 0$, a contradiction. Therefore, for arbitrary $\varepsilon > 0$, there exists $x \in S$ (depending on ε) such that

$$x > u^* - \varepsilon.$$

Conversely, suppose that $u^* \in \mathbb{R}$ satisfies (a) and (b). In particular, u^* is an upper bound of S. Suppose u^* is not the supremum of S. Then there exists an upper bound $u < u^*$ of S. Set

$$\varepsilon = u^* - u > 0.$$

Then

$$x \leq u = u^* - \varepsilon$$

for every $x \in S$, contradicting (b). □

Remark 1. Proposition 2.1 implies that if a set has a supremum, even if the supremum is not an element of the set, it can be approximated arbitrarily closely with an element of the set. For example, consider $S = (0, 1) \subset \mathbb{R}$. Then $\sup(S) = 1 \notin S$. For $\varepsilon = 0.001$, one may take $x = 0.9995 \in S$, which is within 0.001 of the supremum.

Remark 2. A similar ε-characterization holds for the infimum of a nonempty set $S \subseteq \mathbb{R}$ that is lower bounded. What do you think the corresponding statement should be?

Remark 3 (Nested quantifiers: order matters). When quantifiers are nested, their order is important. Recall that $\forall$ is the *universal quantifier*, meaning "for every," while $\exists$ is the *existential quantifier*, meaning "there exists."
In general,

$$\forall x\, \exists y\ P(x, y) \qquad \text{and} \qquad \exists y\, \forall x\ P(x, y)$$

say very different things.
The statement

$$\forall x\, \exists y\ P(x, y)$$

means that for each x, we may choose a possibly different y such that $P(x, y)$ holds. Thus, the choice of y is allowed to depend on x.
In contrast,

$$\exists y\, \forall x\ P(x, y)$$

means that there must be one fixed y that works simultaneously for every x.
For example, in a field F,

$$\forall a \in F\ \exists b \in F \quad a + b = 0$$

is true: for each a, we may choose its additive inverse $b = -a$.

In contrast,

$$\exists b \in F\ \forall a \in F \quad a + b = 0$$

is false for every field, because it would require one single b to equal $-a$ for every a.
Let us look at another example. Let

$$S = \{\text{Alice}, \text{Bob}, \text{Charlie}\},$$

and let $P(x, y)$ mean "x has spoken to y." Then

$$\forall x \in S\ \exists y \in S \quad P(x, y)$$

means that everyone has spoken to someone, whereas

$$\exists y \in S\ \forall x \in S \quad P(x, y)$$

means that there is one person to whom everyone has spoken.

3 The Archimedean Property

Our first step is to show that $\mathbb{N}$ is not an upper bounded subset of $\mathbb{R}$. We have used this fact often when solving problems, and we finally get to prove it.

Theorem 3.1. The set $\mathbb{N}$ of natural numbers is not an upper bounded subset of $\mathbb{R}$.

Proof. Suppose, for the sake of contradiction, that $\mathbb{N}$ is upper bounded. We have that $\mathbb{N}$ is nonempty since $1 \in \mathbb{N}$. By the completeness axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$. Let $\beta = \sup \mathbb{N}$. Then $\beta - 1 < \beta$, so $\beta - 1$ is not an upper bound of $\mathbb{N}$. Thus there exists $n \in \mathbb{N}$ such that

$$n > \beta - 1.$$

It follows that

$$n + 1 > \beta,$$

a contradiction, since $n + 1 \in \mathbb{N}$ and β is an upper bound of $\mathbb{N}$. □

A useful equivalent formulation is the Archimedean property.

Theorem 3.2 (Archimedean property of the real numbers). Let ε and a be real numbers with $\varepsilon > 0$. There exists $n \in \mathbb{N}$ such that

$$n\varepsilon > a.$$

Proof. By the preceding theorem, $\frac{a}{\varepsilon}$ is not an upper bound of $\mathbb{N}$. Therefore, there exists $n \in \mathbb{N}$ such that

$$n > \frac{a}{\varepsilon}.$$

Since $\varepsilon > 0$, it follows that

$$n\varepsilon > a.$$

□

Can you write the statement of the Archimedean property using nested quantifiers?
An especially useful consequence is that reciprocals of natural numbers can be made arbitrarily small.

Theorem 3.3. Let $\varepsilon > 0$. There exists $n \in \mathbb{N}$ such that

$$0 < \frac{1}{n} < \varepsilon.$$

Proof. □

4 The structure of an interval on the real line, and the density of rationals

Let $x, y \in \mathbb{R}$ with $x < y$. We know that the open interval (x, y) is nonempty. Indeed, we have the midpoint

$$x < \frac{x+y}{2} < y,$$

so

$$\frac{x+y}{2} \in (x, y).$$

Thus, between any two distinct real numbers there is another real number. This is closely related to the fact that the set of positive real numbers has no least element.

Proposition 4.1. The set

$$\mathbb{R}_{>0} := \{x \in \mathbb{R} : x > 0\}$$

has no least element.

Proof. Suppose for the sake of contradiction that $r \in \mathbb{R}_{>0}$ is the least element. We have $r > 0$, and hence $0 < \frac{r}{2} < r$. Thus $r/2 \in \mathbb{R}_{>0}$ and $r/2 < r$, a contradiction.
Therefore, no element of $\mathbb{R}_{>0}$ can be its least element. □

Equivalently, there is no positive real number that is less than or equal to every positive real number. This observation is often useful when working with statements that hold for every $\varepsilon > 0$.

Exercise 4.1. Let $x, y \in \mathbb{R}$. Suppose that $x < y + \varepsilon$ for every $\varepsilon > 0$. Prove that $x \leq y$.

We now ask a stronger question. Given $x, y \in \mathbb{R}$ with $x < y$, we have seen that it is easy to find a real number strictly between x and y. Can we always find a *rational* number strictly between them?
The answer is yes. More precisely, we will prove that for every $x, y \in \mathbb{R}$ with $x < y$, there exists $q \in \mathbb{Q}$ such that

$$x < q < y.$$

This property is called the *density of* $\mathbb{Q}$ *in* $\mathbb{R}$. We will prove it in this section.
We will need the following theorem on the existence of the integer part for every real number $x \in \mathbb{R}$. This theorem is of independent interest, and its proof can be found in the supplemental lecture notes.

Theorem 4.1 (Existence of the integer part). For every $x \in \mathbb{R}$ there exists a unique integer $m \in \mathbb{Z}$ such that

$$m \leq x < m + 1.$$

Definition 4.1. The integer m given by the preceding theorem is called the *integer part* of x and is denoted by $[x]$. Thus

$$[x] \in \mathbb{Z}, \qquad [x] \leq x < [x] + 1.$$

Example 4.1. For example,

$$[2.7] = 2, \qquad [-2.7] = -3.$$

We are now ready to prove the density of rationals. The main idea is to start from any interval (x, y) and choose n large enough that the interval (nx, ny) has length greater than 1, and then find an integer in it.

Theorem 4.2 (Density of the rationals). If $x, y \in \mathbb{R}$ and $x < y$, then there exists a rational number q such that

$$x < q < y.$$

Proof. We have $y - x > 0$, and by the Archimedean property with $\varepsilon = y - x$ there exists $n \in \mathbb{N}$ such that

$$n(y - x) > 1,$$

that is,

$$nx + 1 < ny.$$

Then

$$nx < [nx] + 1 \leq nx + 1 < ny.$$

Therefore,

$$x < \frac{[nx] + 1}{n} < y.$$

Since

$$q = \frac{[nx] + 1}{n}$$

is rational, the result follows. □

Question. Are the irrationals also dense on the real line?